

Soroush Khajepour

416-729-7984 | s3khajepour@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Candidate for BAsC in Computer Engineering

Waterloo, ON

Sep 2025 – Apr 2030 (Expected)

TECHNICAL SKILLS

Languages: Ruby, C++, C, C#, Python, JavaScript, TypeScript, HTML, Swift, CSS, SQL

Backend & Frameworks: Node.js, Rails, FastAPI, React, Next.js, PostgreSQL, Redis, Sidekiq, Supabase, REST APIs, WebSockets

DevOps & Tools: AWS (S3), Docker, Vercel, Git, GitHub Actions, Linux, Bash, Postman

Hardware & Embedded: Arduino, STM32, Raspberry Pi, FPGA, Verilog, Quartus, Altium, LTspice, MATLAB

EXPERIENCE

Software Engineering Intern

Roam

Jan 2026 – Apr 2026

Toronto, ON

- Architected and shipped a full-stack **fleet maintenance platform** serving **1,000+** vehicles using **Rails, React**, and **PostgreSQL**, automating maintenance scheduling workflows and reducing manual operations overhead for fleet managers
- Integrated **Geotab** and **Samsara** telematics APIs to stream real-time odometer and vehicle telemetry, eliminating manual data entry for **~70%** of fleet records and accelerating data availability for operations staff
- Optimized **PostgreSQL** schemas with composite indexing and query restructuring, reducing average query latency by **~60%** and improving throughput across high-traffic endpoints
- Engineered event-driven background job infrastructure with **Sidekiq, Redis**, and **AWS S3**, processing **10,000+** async telemetry events daily and improving system reliability under high update volume

Software Developer

UW Reality Labs

Sep 2025 – Dec 2025

Waterloo, ON

- Built **Unity (C#)** VR interaction features for XR prototypes, implementing controller and gesture-based input systems while improving interaction responsiveness and usability during testing
- Refactored XR interaction systems into reusable **C#** components, cutting setup time for new prototypes and improving consistency across controller and gesture-based input handling
- Developed **Python** tooling to log, preprocess, and analyze **~1,000+** spatial tracking and interaction events, integrating real-time sensor streams into **Unity** to improve debugging efficiency by **~30%**

Vice President of Software

Turner Fenton Rocketry Association

Mar 2024 – Jun 2025

Brampton, ON

- Led embedded software development for 2 successful rocket launches, building **Arduino**-based flight control and launch automation systems in **C/C++**
- Engineered firmware for sensor fusion and hardware integration (**IMUs**, switches, servo control), making Turner Fenton the **first high school team** to compete at **Launch Canada**, a national university-level rocketry competition
- Featured in **CBC News** for inspiring youth interest in rocket science and STEM

PROJECTS

Chefit | *Swift, SwiftUI, Node.js, Express, PostgreSQL, OpenAI, Docker, JWT*

Apr 2026

- Built a full-stack **AI-powered** recipe discovery app in **24 hours** at ConHacks 2026 using **SwiftUI** and **Node.js/Express**; integrated **OpenAI** ingredient parsing and parallelized **Edamam** API queries via Swift TaskGroup across **2M+** recipes
- Developed a **recommendation engine** prioritizing recipes by ingredient overlap, expiry date, and cook time using weighted scoring to maximize pantry utilization and reduce food waste
- Containerized backend with **Docker** and **PostgreSQL**, implementing **JWT** auth and **Cloudinary** media storage across **15+** REST endpoints for recipe retrieval, image uploads, and user data management

Waterloo ECE 2025 Class Profile Website | *Next.js, TailwindCSS, Vercel, GitHub Actions*

Jul 2025

- Co-developed and deployed a **Next.js + TailwindCSS** website showcasing **100+** student profiles with dynamic filtering and fully responsive UI optimized for mobile and desktop
- Architected a profile categorization system supporting multiple cohorts, reducing search time for recruiters and peers
- Configured **GitHub Actions** CI/CD pipeline to automate **Vercel** deployments on push, enabling zero-downtime updates and eliminating manual deployment steps for the team